



# COMPUTER SCIENCE AND SOCIETY

## AGENDA

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**Computer Science, Society, and the Information Society**

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**The Role of Data in the Information Society**

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**Economic Impact of the Information Society**

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**Social Impacts of the Information Society**

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**Infrastructure Vulnerability, Computer Science and the Military**

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**Responsibility in Computer Science**

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6

**UNIT 4**

# **SOCIAL IMPACTS OF THE INFORMATION SOCIETY**



## **On completion of this unit, you will be able to...**

- describe the social impact of social networks.
- name the possibilities of surveillance in the digital space.
- identify the opportunities and risks of the digitization of education.
- explain the importance of inclusion and diversity in computer science.
- discuss the role of computer science in relation to sustainability.



1. Why is **diversity** good from a business point of view?
2. How does ICT damage the **environment**?
3. How can ICT be used to **benefit** the environment?

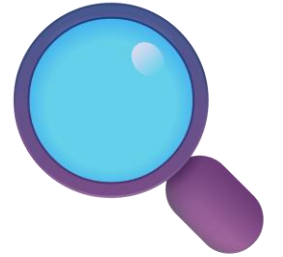
## SOCIAL NETWORKS

- **Facebook: 3.07 billion monthly active users.** Across Meta's apps, there were **3.58 billion daily active people** in December 2025.
- **YouTube** with **more than 3 billion users.**
- **2026:** around **5.79 billion social media “user identities”** worldwide, meaning **more than two in three people on Earth** use social media each month. The typical social media user uses about **6.5 different platforms per month** and spends around **18 hours 36 minutes per week** on social media.

- **Addiction:** Likes and shares result in physical adaptations: brain releases **dopamine**
- Values: e.g. Instagram promoting **narcissism**, projecting **false ideals** of beauty
- **Mobbing**, hate speech, disinformation spread – networks do not do enough to contain it
- Political influence: elections and political **decision-making influenced**. Also: inciting **violence** (see storming of the Capitol)

## SURVEILLANCE

- **Security** reasons cited, e.g. after 9/11 attacks
- **AI systems** learning from human users – **blurry line** to surveillance
- **State surveillance:** e.g. National Security Agency (USA) monitoring private communication





Today:

- States deploying advanced AI surveillance tools to monitor, track, and surveil citizens – e.g. **Pegasus** spyware
- Depends on local laws and values, e.g. system of **social credits** in China would be unacceptable in Europe

**Internet:** created for cooperation between scientists.  
Some successful attempts to bring **education to everyone:**

- OpenCourseWare
- Khan Academy
- Codecademy
- ...

**Future:** digital learning assistants, virtual 3D worlds –  
new possibilities ✨



- Getting **schools online**: hard, even in developed nations
  - **free e-learning** for developing nations: projects often **fail**
    - availability of hardware
    - Lack of technological skills/ **infrastructure** to maintain hardware
    - **social context** of the learners: e.g. safe spaces for learning
- (Ndambakuwa & Brand, 2020).

- McKinsey: Companies with diverse workforce are **more successful** (Dixon-Fyle et al., 2020)
- **Categories** of diversity:
  - gender
  - ethnicity
  - worldview/religion
  - age/generation
  - disability
  - sexual identity/sexual orientation



- Women under represented in CS, but **early pioneers were often women!**
- **Ada Lovelace** (1815-1852), first programmer
- Margaret Hamilton (*see image*, 1902-1985) development of the software for the Apollo programme (**Moon landing**)
- + **Black women mathematicians** and programmers, e.g. Annie Easley: *Hidden Figures* (Melfi et al., 2017).





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**SESSION 3**

# **TRANSFER TASK**

## The problem

- Employers in your region are having **difficulties finding enough qualified computer scientists**.
- You have noticed that computer science jobs are mainly given to **young men with a** higher education background, who were **born in your region**.
- You are trying to get a **more diverse set of people** interested in computer science.
- You decided the best way to go about it is to **focus on kids**.
- In your experience, **parents are not excited** about computer science, whereas many kids think it is **not for them**.



## TRANSFER TASK

### Your Task

1. Create **a curriculum for a one-week course** for 10-year-old kids, aimed at getting them **excited about computer science!**
2. Also create a marketing strategy aimed at reaching as many (and **diverse**) kids as possible and **convincing parents** it is a good idea to for their kids to attend.

Consider...

- What will be the **learning outcomes** in the form of a **verb** (e.g. “After the workshop, kids will be able to **explain** the importance of computer science to society” – except you can come up with something better!) – one or more
- Which **teaching/ learning activities** can you use to address this verb and achieve the outcome? – detailed description of the curriculum
- What **materials** will you/ the kids use? – Literature, Videos, Tools, ...? **Be specific.**
- What **marketing channels** will you use? What will be your **messages?** ...

**TRANSFER TASK**  
**PRESENTATION OF THE RESULTS**

Please present your  
results.

The results will be  
discussed in plenary.





1. Which is not a **social network**?

- a) Facebook
- b) Snapchat
- c) Slack
- d) YouTube



2. Which **spyware** was used by states to spy on their citizens?

- a) Pegasus
- b) Prometheus
- c) Sisyphus
- d) Ikarus



3. Which is **not** a source for free **learning materials**?
- a) OpenCourseWare
  - b) ChatGPT
  - c) Khan Academy
  - d) Codecademy

## LIST OF SOURCES

Dixon-Fyle, S., Dolan, K., Hunt, V., & Prince, S. (2020, May 19). How diversity, equity, and inclusion (DE&I) matter. McKinsey. <https://www.mckinsey.com/featured-insights/diversity-and-inclusion/diversity-wins-how-inclusion-matters>

Draper Laboratory; restored by Adam Cuerden. Public Domain, <https://commons.wikimedia.org/w/index.php?curid=59655977>

Ndambakuwa, S., & Brand, G. (2020, August 13). *Commentary: Many Students in Developing Countries Cannot Access Education Remotely*. The University of Chicago Harris School of Public Policy. <https://harris.uchicago.edu/news-events/news/commentary-many-students-developing-countries-cannot-access-education-remotely>

Woodward, M. (2022). *Social Media User Statistics: How Many People Use Social Media?* <https://www.searchlogistics.com/learn/statistics/social-media-user-statistics/>

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